The Bedrock of Economic Geography

Anders Akerman*

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Abstract

Why do economic maps persist after the advantages that first supported settlement have weakened? I examine this question in Skåne, southern Sweden, linking bedrock maps to several centuries of agricultural and population records and to modern workplace data. Historically, areas with more carbonate-rich bedrock had a larger recorded agricultural base and denser populations. Population records from centuries before industrialization already predict where people live today. The early nineteenth-century population map also predicts where workplaces, employment, and value added are concentrated. The relationship between bedrock and population density was historically present also beyond the surroundings of major cities. Today, it is concentrated around these centers. This contrast suggests that historical agricultural advantages had a lasting impact through the towns they had supported. Their markets, buildings, and transport connections could continue to attract people and firms as employment shifted from agriculture to industry and services.

Keywords: Economic geography; historical persistence; agricultural capacity.
JEL codes: N53, N93, O13, R12.

1 Introduction

Why do economic maps persist after the original productive advantage has faded? Even inside modern high-income countries with common formal institutions, people, workplaces, and production are sharply concentrated. Contemporary concentration is sustained by market access, infrastructure, labor pooling, input-output linkages, and agglomeration economies.

*Stockholm University. Email: anders.akerman@su.se. I thank David von Below, Sophie Cottet, Rikard Forslid, Torfinn Harding, Morten Håvarstein, Thierry Mayer, Dora Simon, Gerhard Toews, Horng Chern Wong, and anonymous referees for helpful comments and suggestions.

These forces can be the channels through which an older economic map, formed when agricultural surplus and settlement depended more directly on the physical landscape, remains economically relevant after the initial advantage has weakened (Krugman, 1991; Ellison and Glaeser, 1999; Head and Mayer, 2004, 2006; Davis and Weinstein, 2002). I study one of the deepest such fundamentals: bedrock.

Bedrock and surface deposits help shape soil properties. Carbonate minerals in rocks such as limestone and marl can buffer soil acidity, improving growing conditions where low pH limits crop production (Goulding, 2016). Together with drainage and topography, these properties influence agricultural capacity and the surplus available to support settlement. These agricultural conditions can in turn shape where villages, churches, roads, and market centers develop. Once a settlement system exists, its usefulness extends beyond the fields that first supported it. Roads connect established places; buildings embody earlier investment; firms gain access to workers and customers. An advantage that first mattered through agriculture can acquire a new economic life through the places built around it.

This paper studies these agricultural origins in Skåne, southern Sweden. I link bedrock maps to historical records of agriculture and population, and then follow the same places into the modern economy. Areas with more carbonate-rich bedrock had a larger recorded agricultural base and denser populations long before industrialization. Historical population still predicts where people live and work, although the modern residential relationship with bedrock is concentrated around major cities. The towns that served the agricultural economy may therefore help explain why its settlement pattern survived structural transformation.

Skåne has considerable geological variation within a small, integrated regional economy. Carbonate-rich formations lie alongside crystalline and other sedimentary rocks. Historical sources record farms, fiscal assessments, population, churches, and towns. The region belonged to Denmark until 1658, when it became part of Sweden, allowing the same landscape to be followed across two systems of administration and recordkeeping. I measure geology by the carbonate-rich share of each historical parish or city area. The main historical outcomes are population density and the recorded agricultural base per square kilometer. Within the same historical boundaries, I measure modern population density in 2025 and mean annual densities of establishments, employment, and value added from private-sector workplace records over 1997–2021.

Carbonate-rich bedrock is strongly and consistently related to the recorded agricultural base and population density. The estimates are positive for both outcomes in all five historical benchmarks from 1570 to 1810, conditional on location and area size. The pooled coefficients remain statistically significant at the 1 percent level with these controls. The population

coefficient is almost identical at the beginning and end of the period, despite Skåne's transfer from Danish to Swedish rule.

Churches, towns, and the broader settlement records are also more concentrated in areas with larger carbonate-rich shares. Across the ten historical town areas in 1810, larger town populations are associated with greater total agricultural capacity in the surrounding countryside. The relationship strengthens as the hinterland is extended from 10 to 30 kilometers, consistent with agricultural districts supporting towns that served a wider market as well as nearby rural communities.

I then compare population density in 1570, 1750, and 1810 as predictors of density in 2025, using the same 381 areas and controls. The 1810 map provides the best single-year fit, but much of its explanatory contribution is already present in 1570. Moving from the twenty-fifth to the seventy-fifth percentile of population density in either 1570 or 1810 corresponds to about 0.44 standard deviations of transformed modern population density, conditional on bedrock, location, and area. The relationship is substantial even for population measured centuries before Sweden's broad industrial transformation.

Including 1810 population density in the regression of modern population on bedrock reduces the modern residential coefficient on bedrock by almost half. Much of the modern relationship with geology is therefore accounted for by the population pattern already established in the agrarian economy. Historical population also strongly predicts the densities of establishments, employment, and value added over 1997–2021, conditional on geology and the geographical controls. Housing or local amenities could keep people in established settlements while they commute elsewhere for work. The employment and production results show that historically populous places also remain centers of economic activity.

The estimates suggest that the spatial pattern of the bedrock relationship has changed. The historical relationships with population and the agricultural base extend beyond the areas surrounding major cities. The modern residential coefficient is close to zero when areas within 10 kilometers of the major city centers are excluded. This contrast suggests that towns supported by the agricultural economy could continue to attract people and firms as employment shifted toward industry and services. Their markets, buildings, transport connections, and labor pools could remain valuable as the surrounding land's economic role changed.

The paper contributes to research on natural advantage and agglomeration (Krugman, 1991; Duranton and Puga, 2004; Rosenthal and Strange, 2004; Puga, 2010; Ellison et al., 2010; Redding and Rossi-Hansberg, 2017). Combes et al. (2010) use geology and historical urban population as instruments for modern employment density when estimating agglomeration effects on wages and productivity. Their geological instruments include limestone

and marl and are motivated in part by agriculture's influence on settlement. I examine the agricultural history behind this persistence using records of farms, mantal, and population across rural parishes and towns. The association between bedrock and population density was already present in 1570 and remained stable through 1810, under both Danish and Swedish rule. The agricultural records document the farming base of these early population concentrations, helping explain why places that remain important today were populous before industrialization.

The agricultural evidence also relates to research on preindustrial population and city formation. Ashraf and Galor (2011) document a positive relationship between land productivity and preindustrial population density. Bosker and Buringh (2017) show how transport geography and proximity to existing cities shaped European city formation. In a dynamic model grounded in historical U.S. evidence, Nagy (2023) explains how agricultural hinterlands support city formation and growth. The Scanian records allow the agricultural base and population to be compared both within parishes and between towns and their surrounding countryside.

Fernihough and O'Rourke (2021) show how industrialization changed the economic importance of geology: European cities near coalfields grew faster after 1750, but not before. I examine a different channel: the persistence of a geography shaped by agriculture before industrialization.

Henderson et al. (2018) find that agricultural geography has greater explanatory power relative to trade geography in countries that developed earlier, despite agriculture's smaller employment share today. They interpret this pattern as a legacy of cities that formed near agricultural production when food transport was costly and persisted as transport costs fell. Their global comparison infers this legacy from contemporary economic geography and the timing of structural transformation. I document its historical basis in Skåne, linking records of agriculture and population in 1570–1810 to the modern distribution of residents and workplaces. Michaels et al. (2012) show how the shift out of agriculture increased population concentration in the United States, with growth shaped by initial density and the agricultural employment share. In Skåne, the historical association of bedrock with agriculture and population extended across the countryside, whereas the modern population relationship is strongest around cities. This pattern suggests that an agricultural geography can persist through the towns it supported, even as employment shifts away from farming.

A further literature examines persistence through major changes in locational advantages. Davis and Weinstein (2002) document the recovery of Japanese cities after wartime destruction. Bleakley and Lin (2012) find persistent population concentrations at obsolete American portage sites, while Jedwab et al. (2017) show that Kenyan railway cities survived

railway decline and the departure of settlers and traders. In Skåne, agriculture's economic role changed gradually. Bedrock remained fixed and farming continued, while established places attracted industry and services. The agricultural assessments help explain why those places were populous in the first place.

Persistence also depends on adaptation. Michaels and Rauch (2018) show that Britain's urban network adjusted more readily to new transport needs after Roman rule than France's; German division and reunification demonstrate the importance of changing market access (Redding and Sturm, 2008; Ahlfeldt et al., 2015). Industrial policy yields contrasting outcomes: manufacturing employment gains in the Tennessee Valley persisted after federal transfers ended (Kline and Moretti, 2014), whereas Chinese counties hosting large industrial plants experienced early productivity gains followed by relative decline (Heblich et al., 2026). These studies distinguish the persistence of established concentrations from their consequences for subsequent growth. Allen and Donaldson (2020, 2022) formalize the roles of contemporaneous and historical spillovers in spatial persistence.

The paper also relates to the broader literature on geography, early technology, and population history in modern development (Olsson and Hibbs, 2005; Nunn, 2009; Comin et al., 2010; Spolaore and Wacziarg, 2013), following these relationships at a local scale within a common modern institutional setting.

The remainder of the paper is organized as follows. Section 2 describes the historical setting and develops the conceptual framework. Sections 3 and 4 present the data and empirical design, respectively. Section 5 examines the links between bedrock, the historical agricultural economy, and settlement. Section 6 studies persistence into the modern economy. Section 7 examines geographical differences in these relationships and assesses spatial dependence. Section 8 concludes.

2 Historical Setting and Conceptual Framework

2.1 The Scanian Landscape

Skåne lies at the southern end of Sweden, facing Denmark across the Öresund (Figure 1). Within this small region, carbonate-rich formations lie alongside other sedimentary rocks, mafic and intermediate rocks, and crystalline landscapes. Plains, intermediate farming districts, and forested areas have different cultivation conditions and settlement histories. Grain production was an important part of the historical economy (Skoglund, 2022); village farming combined crop and animal husbandry, with cultivation and grazing partly organized

collectively (Lagerås and Magnell, 2020). These differences can be studied within a regional economy whose residents and firms share national institutions today.

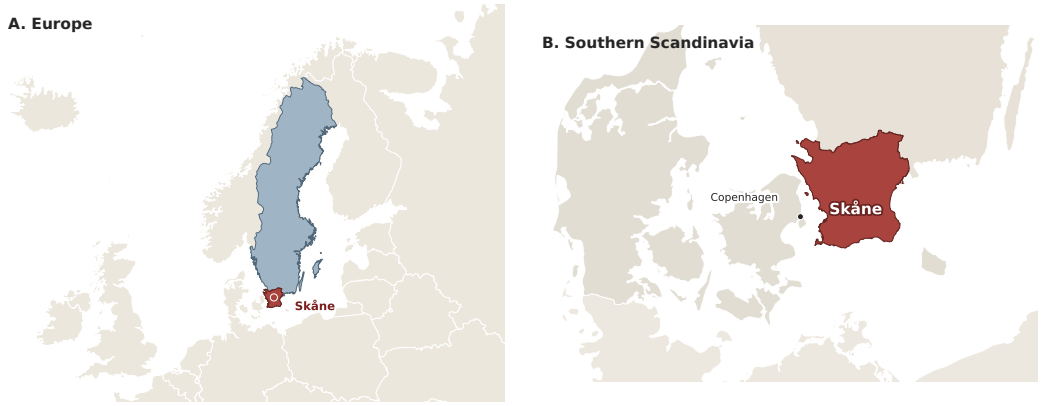


Figure 1: Skåne in northern Europe
Sources: Natural Earth (country boundaries); GADM 4.1 (regional boundary).

Carbonate-rich rocks and sediments affect soil acidity and nutrient availability. Their significance for farming also depends on surface deposits, drainage, terrain, and cultivation practices (Jenny, 1941; Schaetzl and Anderson, 2005; Food and Agriculture Organization of the United Nations, 1973). Bedrock therefore captures broad differences in agricultural conditions, with considerable variation in soil quality and cultivation conditions within each area.

Cultivated soils need not have formed from the bedrock directly beneath them. Surface deposits can alter the connection between bedrock and soil chemistry, while drainage and terrain can make chemically favorable land difficult to farm. The historical assessments and population records show whether geological differences nevertheless correspond to differences in agricultural capacity and settlement over centuries of cultivation.

The records span a major change in state authority. Skåne belonged to Denmark until the Treaty of Roskilde in 1658, when it became part of Sweden. Fiscal and population observations are therefore available from both Danish and Swedish sources. A stable geographical gradient through this transition would indicate continuity in the agrarian economy, although differences in assessment practices and source coverage can affect comparisons across periods, particularly for mantal.

2.2 Agricultural Capacity and Persistent Settlement

An area with greater agricultural capacity can support more farming households and supply nearby settlements with produce. These households also buy goods and services from

traders and craftsmen. Specialized trades need enough customers to cover their costs, and costly travel limits the market available to any one location. A larger agricultural population can therefore sustain more nonfarm activities within reach of its farms. Concentrating these activities in a common center lets households obtain several goods and services in one visit and gives traders access to buyers and suppliers. Repeated use encourages investment in market facilities, buildings, and connecting roads. Towns and villages can in this way grow by serving their agricultural hinterlands. Nagy (2023) formalizes this connection in a model where access to agricultural hinterlands raises the returns to nonfarm production and supports city formation and growth. A town's agricultural base can extend well beyond its boundaries. Its location need not coincide with the most productive land, since access to surrounding households and routes of exchange also matters. Agricultural conditions can thus shape both rural population density and the network of local centers.

Farm counts, mantal assessments, and population describe different parts of this economy. The farm and mantal records indicate the extent of agricultural settlement or its fiscal assessment; population includes both farming households and those supported by trade and services. Their joint relationship with bedrock provides evidence on the agricultural basis of settlement. Landholding and taxation affect the distribution of resources, while migration and family formation affect where people live. All outcomes are measured per unit of land. Favorable agricultural conditions could sustain a larger population at similar living standards.

2.3 Why Established Places Can Remain Valuable

Once a settlement has formed, past investments can make it attractive to new activities. Buildings and transport links are costly to reproduce. A firm entering an established town may find premises, workers, suppliers, and customers already in place. At a new location, some of these advantages would have to be assembled before production could begin. Commercial relationships and familiarity with a market can also remain valuable as the goods being traded change. Firms and households may therefore continue to choose established locations after agriculture has become a smaller part of the economy. In 2023, agriculture, forestry, and fishing together accounted for only 1.5 percent of employment among 16–74-year-olds working in Skåne (Region Skåne, 2026). Current agglomeration economies can reinforce these advantages through labor-market matching, specialized suppliers, and access to customers (Duranton and Puga, 2004; Rosenthal and Strange, 2004; Puga, 2010). Allen and Donaldson (2020, 2022) develop this distinction between contemporaneous and intertemporal forces in spatial persistence.

Structural transformation can also change where this persistence is visible. Farming requires workers to be close to cultivated land, while many industrial and service activities benefit from proximity to other firms and workers. As employment shifts into these activities, fewer households need to live near the fields. If new jobs concentrate in established towns, an agricultural landscape that once supported dispersed rural settlement can remain important through a smaller number of centers. Their continued importance depends on what they offer to new activities: new transport routes or technologies may favor other locations.

2.4 Implications for the Empirical Comparisons

This argument has four implications for the empirical analysis. First, where carbonate-rich bedrock improves cultivation conditions, it should be associated with both a larger agricultural base and higher population density. A population gradient without corresponding evidence in the agricultural records would give weaker support to an agricultural explanation of settlement. I examine farm counts and mantal assessments alongside population, controlling for broad location and area, and compare their relationships with bedrock at each historical benchmark.

Second, agricultural districts with greater capacity can support more people engaged in trade, crafts, and other specialized activities. Town populations should therefore be positively associated with agricultural capacity in the surrounding countryside. The relationship with bedrock inside a town's own boundaries may be weaker, since suppliers and customers can live farther away. I examine both the association of settlement records with bedrock and the relationship between town size and rural capacity at several distances.

Third, if the assets and connections of established places remain useful, historical population density should predict both modern residence and workplace activity, conditional on geology and the geographical controls. Persistence in residence alone could reflect housing or amenities. A relationship with workplace activity would show that historically populous places also remain important for production. Establishments, employment, and value added capture different margins: employment depends on both establishment numbers and size, while value added also reflects industry composition and production per worker. Comparing several historical years shows how far back this relationship can be traced. Differences in fit can reflect subsequent development as well as the quality of the records.

Fourth, if nonfarm activity expands in established centers serving agricultural hinterlands, the relationship with bedrock may become more concentrated around towns. Historically, capacity and population could vary with bedrock across rural areas; today, the

corresponding population relationship may be strongest around cities. I estimate the historical and modern relationships outside the same city neighborhoods to examine this possibility. Such a contrast would be consistent with towns remaining attractive because of their buildings, markets, and connections as agriculture declined in importance.

3 Data

I combine geological maps with historical population and land-based tax records, settlement registers, and confidential modern registry data on residents and workplaces. I measure each within the same historical parish and city boundaries, so the comparisons follow the same places over time. Appendix Table A1 summarizes the sources and variable definitions.

3.1 Geology and Historical Parishes

The bedrock maps come from the Geological Survey of Sweden (SGU). I classify limestone and marl as carbonate-rich bedrock, following SGU's rock-type labels (Geological Survey of Sweden, 2024), and calculate the share of each historical parish or city that is underlain by these rocks. The other categories include mafic and intermediate rocks, other sedimentary rocks, and granite and gneiss. Appendix Figure A1 shows the exact geological map.

The share distinguishes areas that have the same dominant rock type but different amounts of carbonate-rich bedrock. Across the 381 areas used in the regressions, the mean is 0.36, and the 25th and 75th percentiles are zero and 0.98. Much of the variation is between places with little carbonate-rich bedrock and places where it underlies almost the entire area.

Historical observations come from the Sweden 1570–1810 parish panel, available through the Swedish National Data Service (Andersson Palm, 2025). It brings together information on population, agriculture, land ownership, and tax assessments around 1570, 1630, 1690, 1750, and 1810. Even the final population benchmark, in 1810, predates Sweden's broad industrial transformation in the second half of the nineteenth century (Ericsson and Molinder, 2020, p. 817). I match the historical records to parish and city boundaries from Lantmäteriet, Sweden's mapping, cadastral and land registration authority, using place names and identifiers. Where several records refer to the same area, I sum their population and agricultural values before calculating density.

The panel records population in 381 parish and city units at five historical benchmarks, giving 1,905 observations. Agricultural measures are available for 1,894 of these area-year observations. All 381 areas also have modern data on population density and employed-resident density, so comparisons across historical years use the same sample.

I keep cities separate from neighboring rural parishes to distinguish urban centers from their agricultural surroundings. Their population and agricultural capacity can differ sharply, even though rural residents may use markets, services, or workplaces in the nearby city.

The agricultural series combines farm counts for 1570 with later mantal assessments. The 1570 measure (*gårdetal*) counts holdings equally, regardless of size. In Skåne, it is based mainly on the number of tithe payers in Lund diocese's land register (Andersson Palm, 2013, pp. 18–19, 52–53, 131). Mantal records the assessed taxable capacity of agricultural holdings. I refer to these measures collectively as the recorded agricultural base.

The 1750 and 1810 files use the same mantal assessments from af Forsell's 1834 compilation of the land registers of 1825. The source authors apply these assessments to earlier years because mantal had remained broadly stable (Linde, 2012; Linde and Andersson Palm, 2014).

I retain the published values, sum records assigned to the same historical area, and divide by its full area in square kilometers. This harmonizes geography, not the historical units: farm counts are not converted into mantal, nor are earlier assessments rebased to later ones. The agricultural regressions therefore describe the concentration of recorded holdings or assessments within each year. Differences between years need not represent changes in agricultural capacity. The county comparisons in Section 7 concern the later fiscal measures.

The early population figures are reconstructed from household records and assumptions about household size (Andersson Palm, 2013, p. 45). I retain the year labels used in the source database, although some figures refer to nearby years. Population in 1630 is linearly interpolated between estimates around 1620 and 1699 (Andersson Palm, 2012, pp. 7, 147). The population figures for 1750 and 1810 come mainly from official statistics compiled from reports submitted by parish clergy, a system known as Tabellverket (Linde, 2012; Linde and Andersson Palm, 2014).

3.2 Settlement Records

Table 4 uses two measures of recorded settlement features per square kilometer. The first combines religious-building records from Bebyggelseregistret with an indicator for each of ten historical towns. A building enters if its earliest recorded historical event or building phase begins in 1810 or earlier. Within the 381 areas, this gives 341 church records, two bell-tower records, and ten town indicators. A church and a town at the same location each contribute one entry. This measure locates early religious and urban centers.

The building dates describe the history recorded for a building or site, not necessarily the foundation of the surrounding settlement. Town indicators identify the historical city areas listed in Appendix A.1; Skanör and Falsterbo form one combined area.

The broader measure adds 20,123 mapped heritage features from the Swedish National Heritage Board's Fornsök register and 5,129 selected names from Lantmäteriet's Ortnamn database, giving 25,605 entries in total. I select heritage features by their site-type and feature descriptions. They include occupation sites, village and farm plots, house foundations, hearths, cooking pits, church and monastery remains, mills, old routes, and rock carvings with cup marks. The selected names include churches and buildings or settlements with names associated with villages, farms, and estates. I include these features and names regardless of whether they can be dated to before 1810.

For each measure, I count entries within the historical parish or city and divide by its full area in square kilometers. Each entry has equal weight, irrespective of its size. These counts need not correspond to distinct settlements: a heritage record can contain several mapped features, and different sources can describe the same place. Appendix A.1 gives the categories and selection rules. The broader measure covers traces of habitation and local activity beyond the religious and urban centers in the first measure.

For a separate comparison of towns and their agricultural hinterlands, I use the ten historical town areas with population observations in 1810. I locate their cores at central church sites documented before 1810, using coordinates from Bebyggelseregistret. Around each core, I sum rural mantal within 10, 20, and 30 kilometers, allocating each rural parish's total in proportion to the share of its area within the circle. All city areas are excluded. Skanör and Falsterbo have a combined population observation, so I combine the circles around their two church sites, counting overlapping land once. The measure covers rural parishes within the Skåne study region and allocates each parish's mantal uniformly across its area; small areas without mantal observations are omitted.

3.3 Modern Residents and Workplaces

Statistics Sweden's small-area statistics provide population in 2025 and employed residents aged 20–64 in 2024 (Statistics Sweden, 2026a, 2025b). The data cover 819 demographic statistical areas, known as DeSO, using the 2025 boundaries for Skåne (Statistics Sweden, 2025a). I allocate their population and employed-resident counts to the historical areas in proportion to the geographical overlap, then divide by the full area of each historical parish or city.[1] For the maps in Figure 2, I use population in 2021 to match the final year of the

[1]Statistics Sweden makes small adjustments to the published population counts to protect confidentiality. Published components and totals can therefore differ slightly.

workplace records, allocating it from Skåne's 789 DeSO areas on the 2018 boundaries using the same method (Statistics Sweden, 2026a).[2]

Employed-resident density measures where working people live; the workplace records below measure where jobs are located. Commuting separates the two. I use both to examine whether the historical population map predicts where people live and where they work.

Statistics Sweden provides restricted-access plant-level private-sector workplace records covering 1997–2021 (Statistics Sweden, 2026b). I use workplace counts, reported employment, and reported value added to measure economic activity per square kilometer. Recorded coordinates locate each workplace within a one-kilometer grid cell; each cell is assigned to the historical parish or city covering the largest part of its area. This approximation can affect assignments close to parish boundaries. The main results cover 1997–2021; the appendix also reports results for 2012–2021 and 2021, together with additional outcomes.

I sum workplace counts, employment, and value added within each historical area and year, divide by its area in square kilometers, and average the annual densities over the years represented in each period. The regression uses the transformed average, with one observation per parish or city. The full-period average describes private activity over several decades; the shorter-period and single-year results show its more recent distribution.

Workplace density counts establishments per square kilometer, while employment density also reflects their size. Value-added density measures the concentration of production and depends on employment, production per worker, and industry composition. I use value added as reported for each workplace and year.

Figure 2 maps bedrock, population density in 1810 and 2021, and private workplace density in 2021 within the same historical boundaries. In 1810, southern and western Skåne were relatively densely populated, while the northern interior was more sparsely settled. Carbonate-rich bedrock is widespread in the south, although dense settlement also occurs on other formations along the western coast. Much of this broad contrast remains in the modern population map, with more pronounced concentrations along the western coast and around several inland centers. Workplaces are also concentrated along the western coast and in the southwest. The regressions examine these historical and modern patterns conditional on location and the size of each area.

[2] A historical parish covering one quarter of a DeSO area receives one quarter of its reported population and employed residents. This assumes an even distribution of residents within each DeSO area. Concentrated settlement can lead to people being assigned to the wrong side of a historical boundary. The small size of the statistical areas limits this approximation but does not eliminate it.

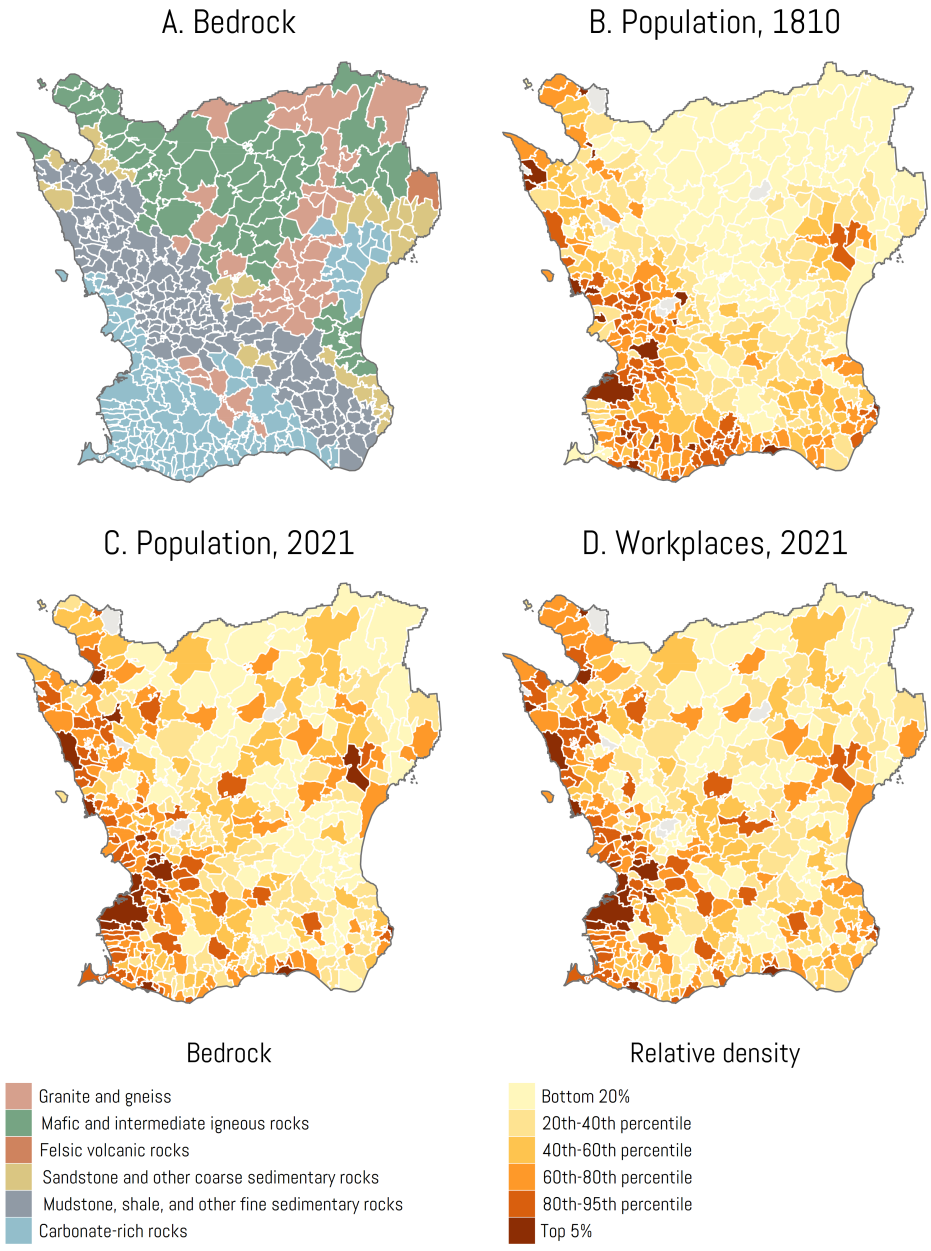


Figure 2: Bedrock and the geography of population and workplaces

Notes: Panel A shows the dominant bedrock group in each historical area. Panels B and C show population density in 1810 and 2021; Panel D shows private workplace density in 2021. Densities use the full historical area in square kilometers. Panels B–D rank the same 381 areas, with equal weight per area and separate 20th, 40th, 60th, 80th, and 95th percentile cutoffs. The shared legend shows relative position within each panel, not equal absolute density. Gray denotes areas not shown. Sources: SGU (bedrock), Sweden 1570–1810 through SND (historical population), Statistics Sweden (modern population and workplaces), and Lantmäteriet (historical boundaries). The coastline follows GADM 4.1.

Table 1: Summary statistics

Variable	Mean	SD	25th pct.	75th pct.	N
historical parish-year panel					
Historical population density	3.436	0.595	3.091	3.806	1,905
Historical farm/mantal density	0.903	0.440	0.569	1.188	1,894
matched parish cross-section					
Carbonate-rich bedrock share	0.363	0.447	0.000	0.980	381
Population density, 1570	3.088	0.589	2.741	3.491	381
Population density, 1750	3.603	0.454	3.389	3.876	381
Population density, 1810	3.973	0.461	3.698	4.248	381
Modern population density	4.450	1.338	3.455	5.349	381
Employed-resident density	3.650	1.329	2.674	4.566	381
Church and town record density	0.053	0.053	0.016	0.074	381
Broader settlement record density	1.566	0.473	1.234	1.879	381

Notes: Density variables are reported on the transformed scale used in the regressions, with raw densities measured per square kilometer. The historical panel covers 1570, 1630, 1690, 1750, and 1810; mantal is missing for 11 parish-year observations. The matched cross-section contains 381 parish/city units at every historical year and for modern population and employed residents. Modern population refers to 2025 and employed residents aged 20–64 to 2024. Settlement measures count the entries defined in Section 3.2 per square kilometer; entries need not identify distinct settlements. Carbonate-rich bedrock remains on its original share scale.

4 Empirical Design

The empirical analysis follows the implications in Section 2.4. I first relate carbonate-rich bedrock to the recorded agricultural base and population density. I then examine settlement records and compare town populations with mantal in the surrounding countryside. Finally, I use the historical population maps to predict the location of modern residents and workplaces. All comparisons use the same historical boundaries.

4.1 Historical Agricultural and Population Density

Let G_i denote the carbonate-rich bedrock share of historical area i, and let q_{it} be population or the agricultural measure per square kilometer at date t. The latter uses farm counts in 1570 and mantal in the later source files. I estimate

$$q_{it} = \alpha + \beta G_i + X_i'\gamma + \lambda_t + \varepsilon_{it}, \tag{1}$$

Density variables in the equations use the inverse-hyperbolic-sine transformation of quantities per square kilometer; geological shares are untransformed. Year effects, λ_t, absorb differences

in average density across years. The first specification includes bedrock share and year effects. The preferred specification adds X_i, containing location and area controls.

Because bedrock is fixed over time, β is estimated from differences between areas. The pooled regression imposes a common coefficient across years; separate regressions show the relationship in each year. Population regressions use the same 381 areas in every year, while agricultural regressions use the available observations.

The location controls are a full cubic polynomial in the coordinates of a representative point inside each historical area.[3] They absorb broad geographical differences in density, including the contrast between northern and southern Skåne. The bedrock coefficient uses the variation remaining after these regional patterns are accounted for. I also control for the area of each parish or city.

The coefficient β is the fitted difference in transformed density between areas with zero and complete carbonate-rich coverage; a ten-percentage-point difference corresponds to 0.1β. With a single geological share in the regression, the comparison is with the mix of other bedrock types found in Skåne.

Bedrock predates the historical economy, ruling out reverse causality from agriculture and settlement to geological composition. Flexible location controls and area size account for broad geographical differences. County controls and exclusions of areas around major cities further examine whether the historical relationship persists within administrative divisions and beyond urban centers.

4.2 Settlement and Modern Economic Activity

For the settlement records, I estimate

$$s_i = \alpha + \beta G_i + u_i. \tag{2}$$

Here s_i is settlement-record density. I estimate the equation separately for church and town records and for the broader measure, without location or area controls.

Towns could draw on agricultural areas beyond their own boundaries. I compare total town population in 1810 with rural agricultural capacity within 10, 20, and 30 kilometers of each historical core. Scatter plots and unweighted linear fits show how the relationship changes as the hinterland expands. Both population and mantal enter in levels.

[3] Coordinates are measured in kilometers and centered before constructing the polynomial. It includes x, y, x^2, xy, y^2, x^3, x^2y, xy^2, and y^3.

I next examine whether places that were densely settled in the past also concentrate people and economic activity today. For each modern density outcome y_i, I estimate

$$y_i = \alpha + \beta_\tau G_i + \theta_\tau q_{i,\tau} + X_i'\gamma + v_i, \tag{3}$$

The main outcomes are population, workplaces, employment, and value added per square kilometer. For modern population, the historical year τ is 1570, 1750, or 1810; the workplace outcomes use 1810. The geographical sensitivity checks also include employed-resident density. For each outcome, I first estimate the regression without historical population, then add it to examine how the bedrock coefficient changes.

The modern population regressions compare the explanatory power of density in 1570, 1750, and 1810 on the same 381 areas with the same geographical controls. I then include all three historical years together and omit one or both earlier years to assess how much they add beyond the 1810 map.

A positive θ_τ means that historically more densely populated areas also have more residents or economic activity per square kilometer today, conditional on geology and the geographical controls. Roads, markets, and institutions developed alongside settlement and may help explain this persistence.

Adding historical population also changes the interpretation of the geological coefficient. Let b_0 be its coefficient without historical population and b_1 its coefficient when $q_{i,\tau}$ is included. Let π_τ be the coefficient on bedrock when that historical population measure is regressed on bedrock and the same controls. With the same observations and controls in all three regressions, ordinary least squares gives

$$b_0 - b_1 = \theta_\tau \pi_\tau. \tag{4}$$

The difference $b_0 - b_1$ is the reduction in the bedrock coefficient when historical population is included.

Agricultural conditions may, for example, have supported both a town and the roads serving it. Those roads could have attracted residents in 1810 and firms later on. Historical population would capture part of the transport advantage, while some could remain correlated with geology after population is included. Measurement error in historical population could also leave a residual bedrock coefficient. Assigning the coefficient change to a particular causal channel therefore requires additional assumptions.[4]

[4] Kwon and Roth (2026) develop tests of full mediation under explicit assumptions about treatment assignment and the relationship between treatment, the mediator, and the outcome. Applying such tests here would require a model of geological exposure and historical population beyond the regressions reported.

4.3 Magnitudes and Inference

Coefficients on transformed densities are not generally elasticities, especially at low densities. For modern population, I therefore report magnitudes on a common scale. Changes in R^2 measure the additional explanatory power of historical population.

I report standardized coefficients and interquartile contrasts for bedrock share and historical population density, both expressed in standard deviations of the dependent variable. All calculations use the variables as entered in the regressions and the common sample of 381 areas.

I cluster the historical-panel standard errors by parish or city. Cross-sectional regressions use heteroskedasticity-robust standard errors. The appendix reports spatial HAC standard errors at several distance cutoffs for the baseline historical and modern residential regressions. These allow errors to be correlated between nearby areas; the panel clustering allows correlation across years within an area. The spatial checks do not cover the settlement regressions, models including all three historical population measures, specifications with county controls, or workplace regressions. The appendix also compares the pooled historical coefficient with estimates obtained by randomly reassigning bedrock shares across parishes.

5 Bedrock and the Historical Economy

If carbonate-rich bedrock favored agriculture, areas with more of it should have supported a larger agricultural base and denser populations. I first examine the population, farm-count, and mantal records, then turn to churches, towns, and other settlements.

5.1 Population and the Agricultural Base

Table 2 reports estimates of equation (1) on the pooled observations from the five historical years. Columns (1) and (3) include year effects; columns (2) and (4) add the cubic coordinate polynomial and area controls.

Carbonate-rich areas were substantially more densely populated, as columns (1) and (2) show. The coefficient is 0.483 in column (1). With the full cubic coordinate polynomial and area controls in column (2), it is 0.179 (standard error 0.060), significant at the 1 percent level.

Columns (3) and (4) show a similarly strong relationship with the recorded agricultural base, measured by farm counts in 1570 and mantal in later years, each per square kilometer. The coefficient on carbonate-rich share is 0.394 in column (3) and 0.141 (standard error 0.043) with the same location and area controls in column (4), significant at the 1 percent

level. This pooled coefficient summarizes the relationship across the source measures; it is not an effect on a common unit of agricultural capacity.

Table 2: Bedrock, population, and the agricultural base, 1570–1810

Dependent variable:	Population density		Agricultural base (farm/mantal density)	
	(1)	(2)	(3)	(4)
Carbonate-rich bedrock share	0.483***	0.179***	0.394***	0.141***
	(0.045)	(0.060)	(0.034)	(0.043)
Observations	1,905	1,905	1,894	1,894
Historical areas	381	381	381	381
R^2	0.42	0.65	0.36	0.58
Location controls	No	Yes	No	Yes
Area control	No	Yes	No	Yes
Year fixed effects	Yes	Yes	Yes	Yes

Notes: Pooled regressions for 1570, 1630, 1690, 1750, and 1810. The agricultural-base series uses farm counts in 1570 and mantal in later years; their construction differs across years (Section 3). Eleven agricultural-base observations are missing. Density variables are measured per square kilometer and transformed as described in Section 4; bedrock share is on a zero-to-one scale. Location controls are a full cubic coordinate polynomial. Standard errors in parentheses are clustered by historical area.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The farm counts and mantal assessments give an agricultural interpretation to the population pattern. A larger farming population could also sustain demand for trade, religious services, and administration in nearby villages and towns. The later mantal figures reflect assessment practices as well as the agricultural base; the county comparisons in Section 7 examine whether broad administrative differences account for those estimates.

The population relationship is remarkably stable over the 240 years from 1570 to 1810 (Table 3). With the same location and area controls, its coefficient ranges from 0.161 to 0.199 and is significant at the 5 percent level in all five years. The agricultural coefficients are also positive throughout, ranging from 0.120 to 0.167, although their source units differ. In particular, the 1750 and 1810 agricultural regressions reuse the same mantal assessments, with different missing observations. The population regressions use the same 381 areas in each year; the agricultural sample ranges from 375 to 381 areas.

The population coefficient is almost identical at the two endpoints: 0.196 in 1570 and 0.199 in 1810. Carbonate-rich areas were already more densely populated centuries before Sweden's broad industrial transformation. They remained so across Skåne's transfer from Danish to Swedish rule and through the end of the historical record.

Table 3: Bedrock, population, and the agricultural base by year

Benchmark	1570	1630	1690	1750	1810
Panel A. Population density					
Carbonate-rich bedrock share	0.196***	0.163**	0.172***	0.161***	0.199***
	(0.069)	(0.067)	(0.065)	(0.061)	(0.070)
Observations	381	381	381	381	381
R^2	0.61	0.58	0.46	0.48	0.54
Panel B. Agricultural base (farm/mantal density)					
Carbonate-rich bedrock share	0.120*	0.134***	0.145**	0.139***	0.167***
	(0.068)	(0.051)	(0.060)	(0.048)	(0.042)
Observations	381	376	381	381	375
R^2	0.55	0.43	0.43	0.55	0.61
Location controls	Yes	Yes	Yes	Yes	Yes
Area control	Yes	Yes	Yes	Yes	Yes

Notes: Years identify database benchmarks, not necessarily the dates of the underlying records (Section 3). Panel B uses farm counts in 1570, unreduced mantal in 1630, and reduced mantal thereafter. The 1750 and 1810 agricultural observations draw on the same reduced-mantal source, not independent assessments. Population uses the same 381 historical areas throughout; Panel B uses the available observations. Variable definitions and controls follow Table 2. Standard errors in parentheses are heteroskedasticity-robust.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

5.2 Settlement and Durable Places

Church and town records locate religious, commercial, and administrative centers in the agrarian economy. Heritage records and data on place names cover a wider range of settlements and local activities. I examine whether these records are more concentrated in carbonate-rich areas.

Table 4 uses the two measures defined in Section 3.2. Column (1) includes church and bell-tower records with histories extending to 1810 or earlier, together with indicators for ten historical towns. Column (2) adds also heritage features and selected names, covering a wider range of local activity without the same date restriction. Both counts are measured per square kilometer in the same 381 historical areas.

Areas with more carbonate-rich bedrock contain more recorded churches, towns, and other settlement features per square kilometer. The carbonate-rich coefficient is 0.032 (standard error 0.007) for churches and towns and 0.323 (0.053) for the broader measure. Both estimates are statistically significant at the 1 percent level.

Table 4: Bedrock and settlement records

Dependent variable	(1) Church and town record density	(2) Broader settlement record density
Carbonate-rich bedrock share	0.032^{***}	0.323^{***}
	(0.007)	(0.053)
Observations	381	381
R^2	0.07	0.09
Location controls	No	No
Area control	No	No

Notes: Both columns use the same 381 historical areas. Column (1) counts 341 church records and two bell-tower records with recorded histories extending to 1810 or earlier, plus ten historical-town indicators. Column (2) adds 20,123 mapped heritage features and 5,129 selected names, without a common date restriction. Counts are divided by historical area in square kilometers; entries need not identify distinct settlements. Section 3.2 and Appendix A.1 give the definitions and selection rules. Density variables are transformed as described in Section 4; bedrock share is on a zero-to-one scale. Standard errors in parentheses are heteroskedasticity-robust.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

5.3 Agricultural Districts and Population Centers

Agricultural capacity and population need not be concentrated in the same part of a district. A town could have many residents and little farmland, drawing produce and customers from nearby parishes. Its markets, crafts, and services would then depend on the agricultural capacity of its hinterland as well as the land within its own boundaries.

Figure 3 relates population in the ten historical town areas in 1810 to rural mantal within 10, 20, and 30 kilometers of their historical cores.

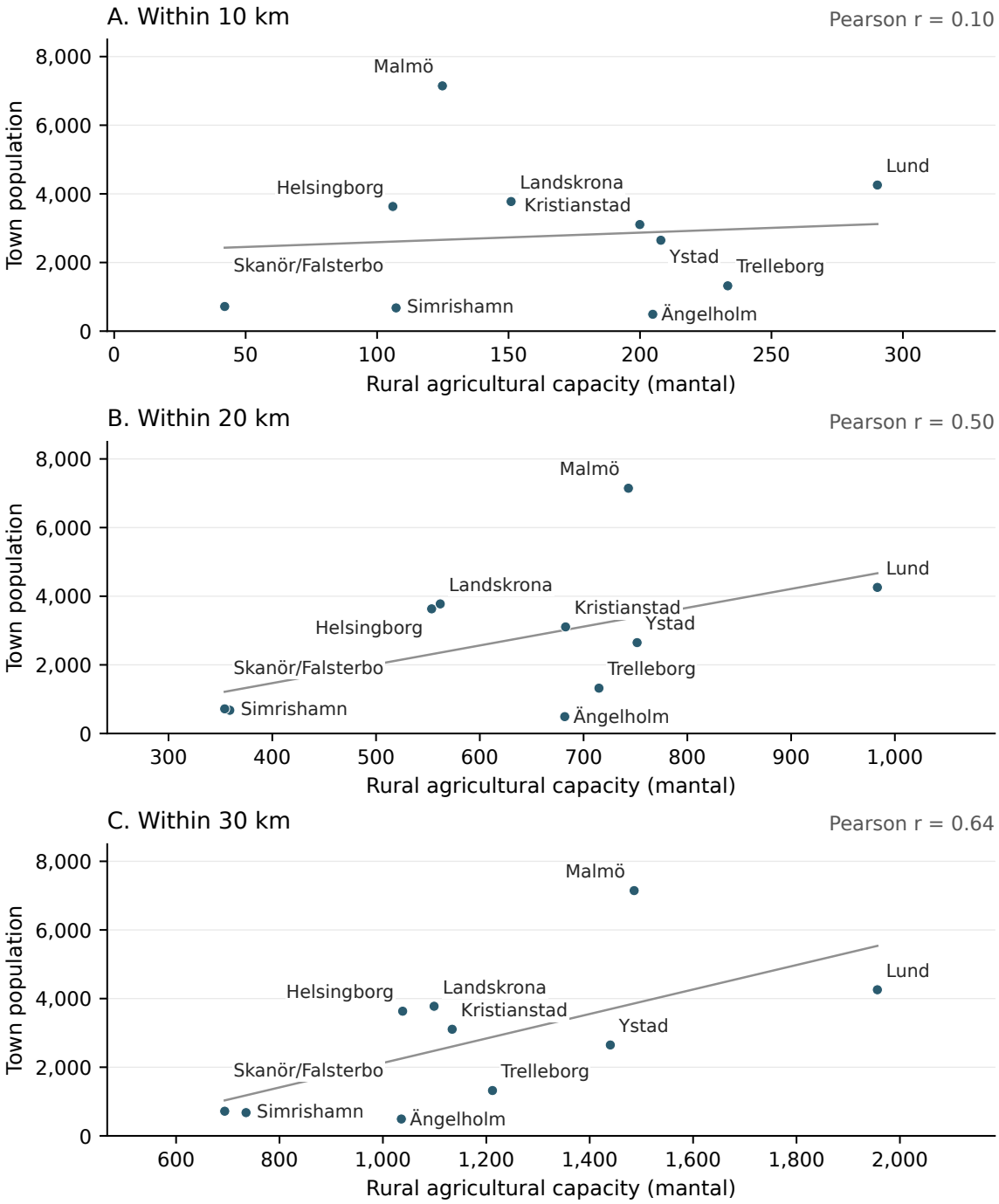


Figure 3: Town population and agricultural hinterlands, 1810

Notes: Each point represents one historical town area. Population is measured around 1810. Mantal comes from the database's 1810 benchmark, based on assessments in the 1825 land registers reported by af Forsell. Rural mantal is summed within the indicated distance of historically documented central church sites, using parish-area overlap weights and excluding all city areas. Skanör and Falsterbo form one observation; their catchments are combined without double counting. The measure covers rural parishes in the Skåne study region. Lines are unweighted linear fits. Sources: Sweden 1570–1810 through SND, Lantmäteriet, and Bebyggelseregistret.

Larger towns tend to have greater assessed agricultural capacity in the wider surrounding countryside. The correlation is 0.10 within 10 kilometers, rising to 0.50 at 20 kilometers and 0.64 at 30 kilometers. The 30-kilometer relationship remains positive when any one town is omitted, with correlations between 0.57 and 0.69.

The positive relationship between town population and agricultural capacity in the surrounding countryside is consistent with the hinterland mechanism developed by Nagy (2023). Total capacity reflects both the extent of rural land and its assessed capacity per unit of area.

The population records include people supported by this wider economy, among them those engaged in trade and services. Buildings, markets, roads, and local services could remain useful as employment shifted toward industry and services.

6 Persistence into the Modern Economy

Carbonate-rich areas had a larger recorded agricultural base and denser populations in the historical economy. Did these population concentrations survive the shift out of agriculture? I examine both modern residents and workplaces, since a historically populous area may remain attractive to households even if employment and production move elsewhere.

6.1 The Historical Population Map

I begin with population in 1810, before Sweden's broad industrial transformation and the expansion of railways, and then ask how far back the relationship with modern residential density can be traced. All comparisons use the same 381 historical areas and the same location and area controls.

In column (1) of Table 5, carbonate-rich share predicts population density in 2025, conditional on location and area. The coefficient is 0.461, significant at the five-percent level. Adding population density in 1810 in column (4) yields a historical-population coefficient of 1.083, significant at the one-percent level.

With population density in 1810 included, the bedrock coefficient falls from 0.461 to 0.245, almost half its original size. Much of the modern relationship with geology is therefore accounted for by where people lived before industrialization. Settlements supported by agriculture may have continued to attract residents because their housing, infrastructure, and market connections retained value as the economy changed.

The size of this decline follows from the historical and modern estimates. Carbonate-rich areas had higher population density in 1810, and historically populous areas remain denser

today. Equation (4) expresses the fall in the bedrock coefficient when population density in 1810 is included as the product of these two coefficients, $0.199 \times 1.083 \approx 0.216$.

Table 5: Historical population and modern residential density

Dependent variable:	Population density, 2025				
	(1)	(2)	(3)	(4)	(5)
Historical population year	None	1570	1750	1810	All three
Carbonate-rich bedrock share	0.461**	0.306	0.302	0.245	0.238
	(0.222)	(0.210)	(0.213)	(0.208)	(0.210)
Population density, 1570		0.791***			0.212
		(0.200)			(0.238)
Population density, 1750			0.984***		0.365
			(0.215)		(0.282)
Population density, 1810				1.083***	0.616**
				(0.225)	(0.296)
Observations	381	381	381	381	381
R^2	0.28	0.33	0.34	0.34	0.35
Adjusted R^2	0.26	0.30	0.31	0.32	0.32
Location controls	Yes	Yes	Yes	Yes	Yes
Area control	Yes	Yes	Yes	Yes	Yes

Notes: All regressions use the same 381 historical areas. Densities are measured per square kilometer and transformed as described in Section 4; bedrock share is on a zero-to-one scale. Location controls are a full cubic coordinate polynomial. Standard errors in parentheses are heteroskedasticity-robust.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The earlier records show that this persistence reaches back to the sixteenth century. Replacing the 1810 measure with population density in 1570 or 1750 gives coefficients of 0.791 and 0.984 in columns (2) and (3), both significant at the one-percent level. Places densely settled in 1570 remain more densely populated today, conditional on bedrock and the geographical controls.

Population in 1570 captures much of the predictive information in the 1810 map. Adding the 1570 measure to the bedrock and geographical controls in column (1) raises R^2 from 0.277 to 0.325, a gain of 4.8 percentage points. Using 1810 raises R^2 to 0.342, a gain of 6.5 percentage points.

Including all three historical years in column (5) adds little to the fit obtained with 1810 alone. The coefficients on population in 1570 and 1750 become smaller and less precise, while the 1810 coefficient remains positive and significant. The historical maps appear to describe much of the same persistent settlement pattern, with 1810 giving the best single-year fit.

6.2 Economic Magnitudes

The implied differences in modern population density are substantial. Table 6 reports standardized coefficients and interquartile contrasts. Moving from the twenty-fifth to the seventy-fifth percentile of population density in either 1570 or 1810 corresponds to about 0.44 standard deviations of modern population density, conditional on bedrock share, location, and area.

The standardized historical-population coefficient is 0.348 for 1570 and 0.373 for 1810, compared with a bedrock coefficient of 0.154 before historical population is included.

Table 6: Standardized associations with modern residential density

Dependent variable:	Population density, 2025	
Increase in predictor:	1 SD	P25–P75
Carbonate-rich bedrock share	0.154	0.338
Population density, 1570	0.348	0.444
Population density, 1750	0.334	0.358
Population density, 1810	0.373	0.445
Observations	381	381

Notes: Entries are fitted differences in standard deviations of the dependent variable, for a one-standard-deviation or interquartile increase in the predictor. Calculations use the variables as entered in Table 5: column (1) for bedrock and columns (2)–(4) for historical population. All specifications control for location and area; historical-population estimates also condition on bedrock share.

To compare the degree of persistence with other settings, I follow the population-density comparison in Davis and Weinstein (2002). Their study of Japan documents regional population patterns that survived centuries of economic change. Between population density in 1600 and 1998, they find a raw correlation of 0.76 and a rank correlation of 0.83.

Across the same 381 historical areas, I correlate historical population density with population density in 2025. The Pearson correlation is 0.48 between 1570 and 2025, 0.46 between 1750 and 2025, and 0.51 between 1810 and 2025. The corresponding Spearman rank correlations are 0.30, 0.23, and 0.34. These are unconditional correlations in untransformed population per square kilometer, with equal weight for each area.

The correlations are lower than in Japan, especially for ranks. The geographical scale is also very different: the Japanese comparison covers 39 large regions across the country, while the Scanian analysis follows 381 smaller areas within a single region.

6.3 Workplaces, Employment, and Value Added

Do historically populous areas also attract establishments, employment, and production? Housing and local amenities may keep people in established settlements even when they commute elsewhere for work. The workplace regressions test whether persistence also extends to the location of economic activity.

Table 7 reports regressions for average workplace, employment, and value-added densities over 1997–2021. Panel A relates these outcomes to carbonate-rich share, the geological measure associated with the historical agricultural base. Conditional on location and area, carbonate-rich share enters positively in all three regressions, with coefficients of 0.317, 0.813, and 1.133. Panel B adds population density in 1810 to ask how much of this relationship can be accounted for by settlement before industrialization. The historical-population coefficients are 0.956 for workplaces, 1.271 for employment, and 1.338 for value added, all significant at the one-percent level.

Table 7: Historical population and modern economic activity, 1997–2021

Dependent variable:	(1) Workplace density	(2) Employment density	(3) Value-added density
Panel A. Without historical population			
Carbonate-rich bedrock share	0.317^{**}	0.813^{***}	1.133^{**}
	(0.160)	(0.232)	(0.469)
Observations	381	381	381
R^2	0.27	0.23	0.12
Panel B. With historical population			
Carbonate-rich bedrock share	0.126	0.560^{***}	0.866^{*}
	(0.141)	(0.209)	(0.469)
Population density, 1810	0.956^{***}	1.271^{***}	1.338^{***}
	(0.171)	(0.269)	(0.275)
Observations	381	381	381
R^2	0.38	0.31	0.15
Location controls	Yes	Yes	Yes
Area control	Yes	Yes	Yes

Notes: Statistics Sweden's establishment records cover the private sector. Workplace density counts establishments per square kilometer; employment and value added are totals per square kilometer. Annual densities are averaged over the years observed within 1997–2021, then transformed as described in Section 4. All regressions use the same 381 historical areas. Location controls are a full cubic coordinate polynomial. Standard errors in parentheses are heteroskedasticity-robust.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

With historical population included, the geological coefficients fall to 0.126, 0.560, and 0.866. The remaining workplace-density estimate is imprecise; employment remains significant at the one-percent level and value added at the ten-percent level. The earlier population pattern thus accounts for part of the concentration of modern activity in carbonate-rich areas.

Employment density depends on both the number and size of establishments, while value-added density also reflects sector composition and production per worker. Established centers may attract larger employers or particular industries through access to workers, suppliers, and transport networks. Population density in 1810 need not capture all of these inherited advantages, and this could help explain why employment and value added remain related to bedrock after historical population is included.

Combes et al. (2010, Table 1.2) document a similar historical-population relationship for modern employment in France. Across 306 employment areas, they report correlations of 0.75 and 0.88 between log urban population density in 1831 and 1881, respectively, and log employment density during 1976–1996. For Skåne, the unconditional Pearson correlation between log population density in 1810 and log mean annual private-sector employment density over 1997–2021 is 0.36, giving equal weight to the same 381 historical areas. The French correlations are higher, but their national comparison uses larger labor-market areas and historical population measured only in towns. The Scanian calculation includes rural populations and compares smaller parish and city areas within a single region.

Appendix Table A2 uses shorter time windows and additional monetary outcomes. The historical-population coefficients remain positive throughout, while the workplace-count coefficient on geology remains imprecise conditional on historical population.

7 Geography and Spatial Inference

The agricultural interpretation raises several geographical questions. County institutions could have affected tax assessments and contributed to the relationship between bedrock and mantal. I examine whether that relationship remains within counties. I then exclude areas near major cities to ask whether the historical relationships with the agricultural base and population extend across the countryside, and whether the modern population relationship is concentrated around cities. I also control specifically for proximity to Copenhagen, since access to this large market could help explain the modern concentration of activity. Finally, shared markets, infrastructure, and physical conditions may correlate the errors of neighboring areas. I examine how allowing for that dependence affects the precision of the estimates.

7.1 County Differences

Skåne was divided into two counties, Malmöhus and Kristianstad. Their institutions could have assessed agricultural holdings differently for taxation. If assessment practices coincided with geology, they could contribute to the estimated bedrock coefficient. Adding county fixed effects to the baseline location and area controls allows me to compare areas under the same county administration.

The two counties' boundaries were broadly stable between 1719 and 1996 (Linde and Andersson Palm, 2014, pp. 9, 83, 89), so I restrict the historical comparison to 1750 and 1810. I use the sources' harmonized county classification, which does not reconstruct the boundary separately for each year.

County fixed effects leave the estimates almost unchanged (Table 8). The population coefficient moves from 0.161 to 0.157 in 1750 and from 0.199 to 0.198 in 1810; the mantal coefficients move from 0.139 to 0.133 and from 0.167 to 0.164. All remain positive and statistically significant. Carbonate-rich areas had denser populations and greater assessed capacity even within the same county.

Columns (3)–(4) exclude eleven parishes whose recorded county affiliation in 1742 is split or differs from the harmonized classification. The estimates change little, so these discrepancies do not explain the results.

The historical-to-modern estimates are similarly stable with county fixed effects (Appendix Table A3). The coefficient on population density in 1810 changes from 1.083 to 1.084 for modern population and from 1.062 to 1.063 for employed residents.

7.2 Cities and Access to Copenhagen

Did the historical agricultural relationship extend beyond the major cities, and does the same hold today? Table 9 reports four comparisons with the baseline: controlling for distance to Copenhagen, excluding areas within 10 kilometers of the six major city centers, excluding historical areas corresponding to towns that existed before 1810, and excluding areas within 30 kilometers of Copenhagen. All retain the baseline location and area controls. Appendix Figure A2 maps the cities, towns, and areas within 10 kilometers of the major city centers.

The historical relationship extends well beyond the surroundings of major cities. Excluding the 10-kilometer neighborhoods in column (3) leaves a population coefficient of 0.159, compared with 0.179 in the full panel, and a farm/mantal coefficient of 0.125, compared with 0.141. Both remain significant at the five-percent level. The separate population regression for 1810 gives the same pattern.

Table 8: Bedrock, historical density, and county fixed effects

	Full sample		Excluding 11 parishes	
	(1)	(2)	(3)	(4)
Panel A. Population density, 1750				
Carbonate-rich bedrock share	0.161***	0.157**	0.169***	0.163***
	(0.061)	(0.062)	(0.061)	(0.063)
Observations	381	381	370	370
R^2	0.48	0.48	0.49	0.49
Panel B. Population density, 1810				
Carbonate-rich bedrock share	0.199***	0.198***	0.202***	0.198***
	(0.070)	(0.070)	(0.071)	(0.072)
Observations	381	381	370	370
R^2	0.54	0.54	0.54	0.54
Panel C. Mantal density, 1750				
Carbonate-rich bedrock share	0.139***	0.133***	0.134***	0.127**
	(0.048)	(0.049)	(0.049)	(0.050)
Observations	381	381	370	370
R^2	0.55	0.56	0.55	0.56
Panel D. Mantal density, 1810				
Carbonate-rich bedrock share	0.167***	0.164***	0.163***	0.159***
	(0.042)	(0.042)	(0.043)	(0.043)
Observations	375	375	364	364
R^2	0.61	0.61	0.61	0.61
Location controls	Yes	Yes	Yes	Yes
Area control	Yes	Yes	Yes	Yes
County fixed effects	No	Yes	No	Yes

Notes: Panel headings identify the dependent variable and year. Location controls are a full cubic coordinate polynomial. County fixed effects distinguish Malmöhus and Kristianstad using the historical sources' harmonized classification. Columns (3)–(4) exclude eleven parishes whose recorded 1742 county affiliation is split or differs from that classification. Six mantal observations are missing in 1810. Standard errors in parentheses are heteroskedasticity-robust.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 9: Bedrock and density: city exclusions and Copenhagen

	Baseline (1)	Copenhagen distance (2)	Outside city neighborhoods (3)	Excluding pre-1810 towns (4)	Outside Copenhagen 30 km (5)
Panel A. Population density, 1570–1810					
Carbonate-rich share	0.179***	0.151**	0.159**	0.115**	0.173***
	(0.060)	(0.065)	(0.065)	(0.048)	(0.062)
Observations	1,905	1,905	1,505	1,855	1,860
R^2	0.65	0.65	0.70	0.71	0.65
Panel B. Agricultural base (farm/mantal density), 1570–1810					
Carbonate-rich share	0.141***	0.097**	0.125**	0.155***	0.160***
	(0.043)	(0.045)	(0.057)	(0.039)	(0.042)
Observations	1,894	1,894	1,498	1,855	1,851
R^2	0.58	0.58	0.57	0.62	0.59
Panel C. Population density, 1810					
Carbonate-rich share	0.199***	0.144**	0.186**	0.124**	0.183***
	(0.070)	(0.071)	(0.082)	(0.055)	(0.066)
Observations	381	381	301	371	372
R^2	0.54	0.54	0.62	0.62	0.55
Panel D. Population density, 2025					
Carbonate-rich share	0.461**	0.367	0.029	0.314	0.493**
	(0.222)	(0.250)	(0.241)	(0.209)	(0.228)
Observations	381	381	301	371	372
R^2	0.28	0.28	0.27	0.29	0.27
Panel E. Employed-resident density, 2024					
Carbonate-rich share	0.470**	0.406*	0.028	0.324	0.507**
	(0.218)	(0.245)	(0.233)	(0.204)	(0.223)
Observations	381	381	301	371	372
R^2	0.30	0.30	0.30	0.32	0.29
Location controls	Yes	Yes	Yes	Yes	Yes
Area control	Yes	Yes	Yes	Yes	Yes

Notes: Entries are coefficients on carbonate-rich bedrock share; panel headings identify the dependent variables. Panels A–B pool the five historical years and include year fixed effects. Location controls are a full cubic coordinate polynomial. Column (2) adds $\log(1+d)$, where d is distance to Copenhagen in kilometers. Columns (3)–(5) exclude, respectively, areas within 10 km of Malmö, Lund, Helsingborg, Kristianstad, Ystad, or Trelleborg; pre-1810 town parishes; and areas within 30 km of Copenhagen. Distances are straight-line distances from each area's representative interior point. Employed residents are aged 20–64. Standard errors in parentheses are clustered by historical area in Panels A–B and heteroskedasticity-robust in Panels C–E. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The modern relationship between bedrock and residential density is concentrated around major cities. Excluding areas within 10 kilometers of the city centers reduces the coefficients from 0.461 to 0.029 for population and from 0.470 to 0.028 for employed residents. Excluding only the ten historical areas corresponding to towns that existed before 1810 leaves larger coefficients of 0.314 and 0.324, respectively. Both sets of estimates are imprecise. The larger decline when excluding the wider surroundings—80 of the 381 areas—nevertheless suggests that the modern relationship extends beyond the old town areas into their surroundings.

Agricultural districts may have sustained towns whose markets, buildings, and transport connections remained valuable as employment shifted toward industry and services. The geographical contrast fits this account: the modern population advantage of carbonate-rich areas is clearest around cities, while the historical relationship extended across the wider countryside.

Access to Copenhagen could also help explain modern concentration. Controlling for distance to Copenhagen reduces the modern bedrock coefficients to 0.367 and 0.406. Excluding the nine areas within 30 kilometers leaves coefficients of 0.493 and 0.507, both significant at the five-percent level. The relationship therefore extends beyond the areas nearest Copenhagen.

7.3 Spatial Dependence

Nearby parishes share physical geography, markets, and historical institutions, which may correlate their regression errors. Parish clustering accounts for dependence across years within an area. To allow for dependence between neighboring areas, Appendix Tables A4 and A5 report spatial HAC standard errors (Conley, 1999) with distance cutoffs of 25, 50, and 100 kilometers. The calculations cover the baseline historical bedrock regressions, pooled and for 1810, and the regressions of modern residential density on population in 1810.

For the pooled historical population coefficient of 0.179, the standard error rises from 0.060 with parish clustering to 0.092 at the 50-kilometer cutoff. The farm/mantal coefficient of 0.141 has a standard error of 0.054 at that distance. Allowing for spatial dependence reduces precision more for historical population than for the agricultural measure. Population in 1810 remains significant at the one-percent level in the modern population regression at all three cutoffs, as it does for employed-resident density.

Appendix Table A4 also compares the estimates with those obtained after randomly reassigning bedrock shares across parishes. I shuffle the observed carbonate-rich shares 999 times, re-estimating the pooled population and agricultural regressions after each shuffle. Outcomes and controls stay with their original parish, and the reassigned bedrock share

stays fixed across years. For both outcomes, fewer than 1 percent of the reassignments produce a coefficient as large in absolute value as the observed estimate.

8 Conclusion

The economic geography of modern Skåne retains a clear link to its agrarian past. Areas with more carbonate-rich bedrock had denser populations and a larger recorded agricultural base: more holdings in the earliest farm counts and higher mantal density in the later fiscal records. The population relationship endured from 1570 to 1810, under both Danish and Swedish rule. The agricultural measures also support this interpretation.

Churches, towns, and other recorded settlement features are also more concentrated in carbonate-rich areas. In 1810, larger towns tended to have greater total agricultural capacity in the wider surrounding countryside, consistent with agricultural districts sustaining both rural communities and nearby centers of trade and services.

Population density in 1570, centuries before Sweden's broad industrial transformation, already predicts modern density. The 1810 population map provides the best single-year fit, but an interquartile increase in either historical measure corresponds to about 0.44 standard deviations in transformed modern population density, conditional on bedrock, location, and area. Much of the persistent population pattern was therefore in place at the beginning of the historical record.

Persistence extends to production. Population density in 1810 predicts establishment, employment, and value-added densities over 1997–2021. Historically populous places still attract economic activity, giving their continued importance an economic basis beyond inherited housing and residential amenities.

The geographical reach of the bedrock relationship has changed. Historically, areas with more carbonate-rich bedrock had a larger recorded agricultural base and denser populations even outside the surroundings of major cities. Today, the relationship between bedrock and population density is concentrated around these cities. Towns supported by the agricultural economy may have remained attractive because of their markets, buildings, transport connections, and labor pools. An agricultural advantage can thus persist through the towns it helped sustain, even as the economic role of the surrounding land changes.

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A Data Sources and Definitions

Table A1: Data sources and measurement

Measure	Source	Measurement
Carbonate-rich bedrock	Geological Survey of Sweden	Share of each historical parish or city underlain by carbonate-rich bedrock.
Historical population and agricultural base, 1570–1810	Swedish National Data Service parish panel	Population and farm/mantal totals aggregated within each historical area and divided by its area.
Church and town records	Bebyggelseregistret and historical towns identified in Lantmäteriet's city boundaries	Religious-building records with histories extending to 1810 or earlier, plus one indicator per historical town, divided by historical area.
Broader settlement records	The preceding sources, Fornsök, and Lantmäteriet Ortnamn	The church and town count plus selected mapped heritage features and names, divided by historical area. Selection rules are given in Appendix A.1.
Modern population and employed residents	Statistics Sweden small-area statistics (DeSO)	Counts allocated according to the share of each DeSO area covered by a historical area, then expressed per square kilometer.
Workplaces, employment, and value added	Statistics Sweden, restricted-access private-sector workplace records	Workplace coordinates identify their historical area. Annual workplace counts, employment, and value added are summed within each area, divided by its area, and averaged over the relevant period.

Notes: Historical parish and city boundaries are from Lantmäteriet. The main matched sample contains 381 areas. The agricultural series uses unweighted farm counts in 1570, unreduced mantal in 1630, and reduced mantal thereafter; see Section 3.

A.1 Settlement records

Table 4 uses two measures: religious-building and town records (353 entries), and a broader measure that adds heritage features and selected names (25,605 entries). Both are expressed per square kilometer of the full parish or city area and cover the same 381 historical areas.

Religious buildings and towns. I select church, chapel, funerary-chapel, and bell-tower records from Bebyggelseregistret if their earliest recorded historical event or building phase

begins by 1810. Keeping one record per building identifier gives 341 churches and two bell towers. These dates describe building histories, not necessarily the founding of settlements.

The ten town areas, defined by Lantmäteriet's city boundaries, are Helsingborg, Kristianstad, Landskrona, Lund, Malmö, Simrishamn, Skanör med Falsterbo, Trelleborg, Ystad, and Ängelholm. Each contributes one town indicator; Skanör and Falsterbo form a single area.

Heritage features. The broader measure adds 20,123 mapped features from Fornsök: occupation sites, village and farm plots, croft remains, house foundations, hearths and cooking pits, religious remains, mills, old routes, and cup-mark rock carvings.

I search the site-type and feature descriptions for *boplats*, *boplatsområde*, *boplatslämning*, *härd*, *kokgrop*, *husgrund, förhistorisk/medeltida*, *grav- och boplatsområde*, *bytomt*, *gårdstomt*, *lägenhetsbebyggelse*, *husgrund, historisk tid*, *kyrka/kapell*, *kloster*, *kvarn*, and *färdväg*. The search ignores capitalization and includes matches within longer descriptions. There is no restriction on age, recording method, or protection status.

Names. Lantmäteriet's Ortnamn database contributes 5,129 entries: church names and selected building or settlement names. Names classified as buildings or settlements qualify if they end in *-by*, *-torp*, *-arp*, *-rup*, *-löv*, *-lev*, *-stad*, *-tofta*, *-inge*, *-åkra*, *-hult*, *-ryd*, *-rum*, *-lunda*, or *-lösa*; contain *gård*, *gården*, or *gard*; or end in *säteri*, *herrgård*, *slott*, *kungsgård*, *sätesgård*, *herresäte*, *gods*, or *borg*. Selection uses names in the modern register without a historical date restriction.

Location and counting. Entries are assigned to the historical area containing their coordinates; lines and polygons are represented by interior points. A feature or name matching several search terms is counted once. The 20,123 heritage features correspond to 19,167 register identifiers, since one record can contain several mapped features. Records from different sources, including church records and town indicators, remain separate even when they describe the same place.

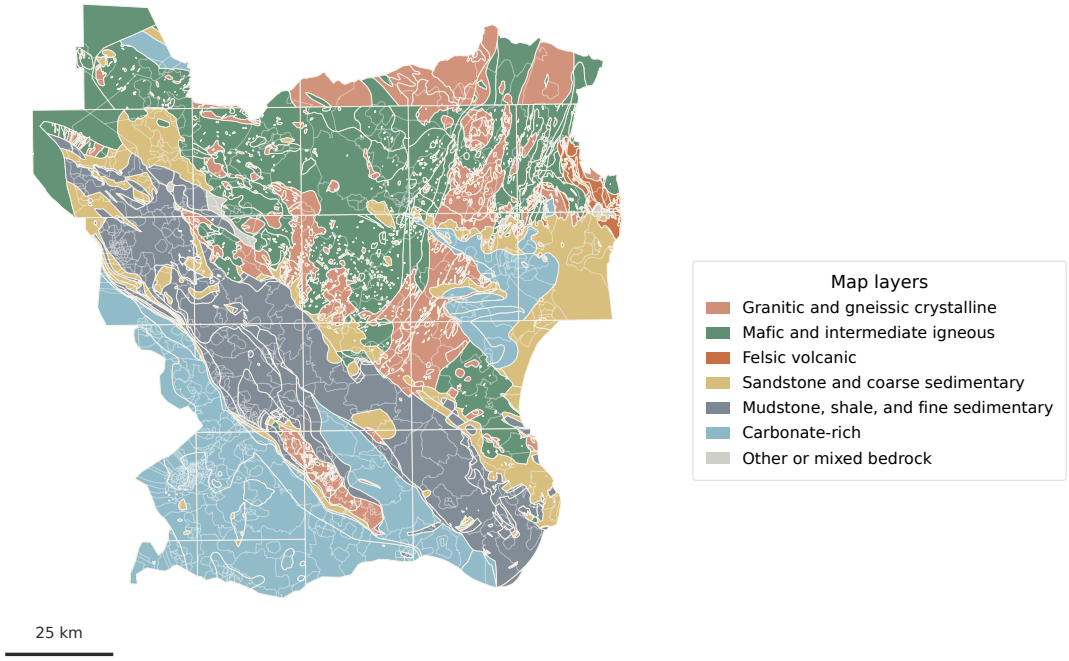


Figure A1: Bedrock geology in Skåne

Notes: Colors show the bedrock groups used in the analysis. Bedrock data are from SGU; the map outline and local boundaries are from Statistics Sweden.

B Modern Economic Activity

Table A2: Modern economic activity across time periods

Dependent variable	Workplaces		Employment		Value added		Production value		Labor costs		Raw-material costs	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Panel A. 1997–2021												
Carbonate-rich bedrock	0.32**	0.13	0.81***	0.56***	1.13**	0.87*	0.69***	0.40*	0.86***	0.57**	0.80***	0.50*
	(0.16)	(0.14)	(0.23)	(0.21)	(0.47)	(0.47)	(0.24)	(0.21)	(0.27)	(0.25)	(0.28)	(0.26)
Population density, 1810		0.96***		1.27***		1.34***		1.42***		1.48***		1.51***
		(0.17)		(0.27)		(0.28)		(0.27)		(0.29)		(0.31)
R^2	0.27	0.38	0.23	0.31	0.12	0.15	0.22	0.31	0.22	0.30	0.19	0.26
Panel B. 2012–2021												
Carbonate-rich bedrock	0.31*	0.12	0.84***	0.58***	1.15**	0.88*	0.67***	0.38*	0.86***	0.56**	0.75**	0.44
	(0.16)	(0.14)	(0.24)	(0.21)	(0.48)	(0.49)	(0.25)	(0.22)	(0.28)	(0.26)	(0.30)	(0.27)
Population density, 1810		0.94***		1.30***		1.36***		1.46***		1.54***		1.56***
		(0.17)		(0.27)		(0.28)		(0.27)		(0.30)		(0.31)
R^2	0.28	0.38	0.24	0.32	0.11	0.15	0.21	0.31	0.22	0.30	0.18	0.26
Panel C. 2021												
Carbonate-rich bedrock	0.34**	0.16	0.95***	0.69***	0.86***	0.58***	0.80***	0.52**	1.03***	0.73***	0.84***	0.54*
	(0.16)	(0.14)	(0.24)	(0.22)	(0.24)	(0.22)	(0.25)	(0.23)	(0.28)	(0.26)	(0.30)	(0.28)
Population density, 1810		0.93***		1.30***		1.42***		1.44***		1.50***		1.53***
		(0.17)		(0.27)		(0.27)		(0.27)		(0.30)		(0.32)
R^2	0.29	0.39	0.25	0.32	0.23	0.32	0.21	0.30	0.23	0.30	0.19	0.26
Geographic controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	381	381	381	381	381	381	381	381	381	381	381	381

Notes: Restricted-access private-sector workplace records from Statistics Sweden. All outcomes are measured per square kilometer. Annual densities are averaged within each multiyear period before transformation. Within each outcome pair, the second column adds population density in 1810. Each regression uses one observation per historical parish or city; the geographic-control and observation rows apply to all panels. Carbonate-rich bedrock is a zero-to-one area share. Geographic controls comprise a full cubic polynomial in coordinates and a separate control for area. Standard errors in parentheses are heteroskedasticity-robust. Stars denote * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

C Geographical Sensitivity

Table A3: Modern residential density and historical county fixed effects

	(1)	(2)	(3)	(4)
Panel A. Dependent variable: population density (2025)				
Carbonate-rich bedrock share	0.46**	0.46**	0.24	0.25
	(0.22)	(0.23)	(0.21)	(0.21)
Population density, 1810			1.08***	1.08***
			(0.22)	(0.23)
R^2	0.28	0.28	0.34	0.34
Panel B. Dependent variable: employed-resident density (2024)				
Carbonate-rich bedrock share	0.47**	0.47**	0.26	0.26
	(0.22)	(0.22)	(0.20)	(0.20)
Population density, 1810			1.06***	1.06***
			(0.22)	(0.22)
R^2	0.30	0.30	0.37	0.37
Location controls	Yes	Yes	Yes	Yes
Area control	Yes	Yes	Yes	Yes
County fixed effects	No	Yes	No	Yes
Observations	381	381	381	381

Notes: Both panels use the same 381 historical areas. Employed residents are aged 20–64. Location controls are a full cubic coordinate polynomial. County fixed effects use the harmonized Malmöhus and Kristianstad classification described in Table 8. Standard errors in parentheses are heteroskedasticity-robust.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

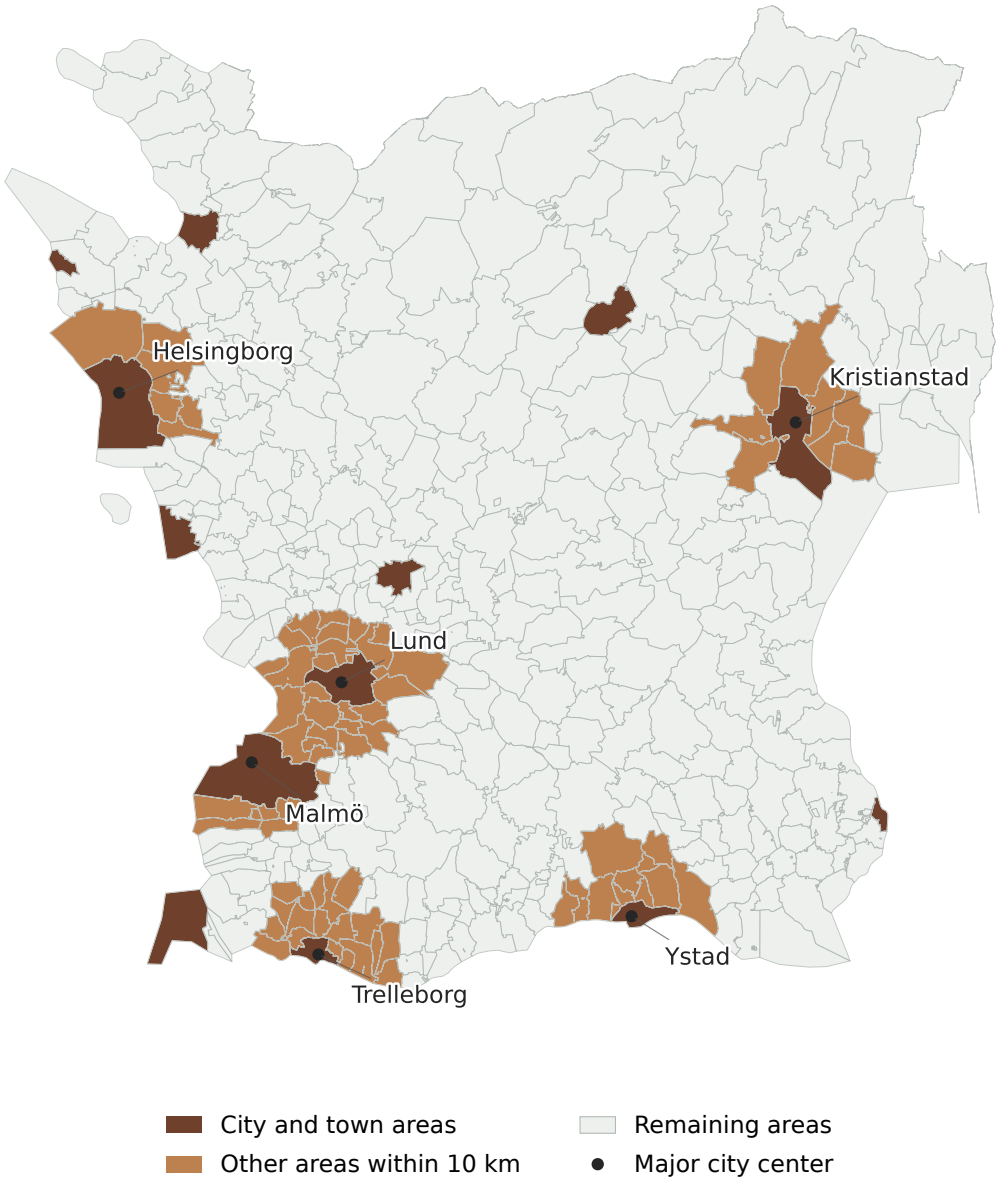


Figure A2: Cities, towns, and major-city neighborhoods

Notes: Brown areas comprise city units in the boundary layer, units classified as pre-1810 towns, and the six major-city parishes. Orange marks other areas whose representative interior point lies within 10 kilometers of a marked city center. Whole areas are shaded. Distances are straight-line distances in SWEREF 99 TM. Boundaries are from Lantmäteriet; town classifications use the historical-town layer.

Table A4: Spatial inference for historical population and the agricultural base

Dependent variable	Coefficient	Parish-clustered standard error	Spatial standard errors (Conley) 25 km	50 km	100 km	Permutation p-value	Observations
Population density	0.18	0.06	0.09	0.09	0.06	< 0.01	1,905
Farm/mantal density	0.14	0.04	0.05	0.05	0.04	< 0.01	1,894

Notes: Coefficients are on carbonate-rich bedrock share in the pooled regressions of Table 2, with location and area controls and year fixed effects. Both regressions cover 381 historical areas. Spatial standard errors use distances between representative interior points and Bartlett weights with the indicated cutoffs, allowing correlation across areas and years. Two-sided permutation p-values compare the observed coefficients with 999 random reassignments of bedrock shares across areas. Each assigned share is held fixed across years; outcomes and controls are unchanged. The reassignments do not preserve spatial correlation in geology.

Table A5: Spatial inference for historical and modern density

Dependent variable	Coefficient	Heteroskedasticity-robust standard error	Spatial standard errors (Conley) 25 km	50 km	100 km	Observations
Panel A. Coefficient on carbonate-rich bedrock share						
Historical population density	0.18	0.03	0.09	0.09	0.06	1,905
Farm/mantal density	0.14	0.03	0.05	0.05	0.04	1,894
Population density, 1810	0.20	0.07	0.10	0.10	0.07	381
Panel B. Coefficient on population density in 1810						
Population density, 2025	1.08	0.22	0.23	0.22	0.20	381
Employed-resident density, 2024	1.06	0.22	0.22	0.21	0.19	381

Notes: All regressions include the baseline location and area controls. The first two rows of Panel A also include year fixed effects; Panel B controls for carbonate-rich bedrock share. Employed residents are aged 20–64. The heteroskedasticity-robust standard errors omit the finite-sample correction used in the main cross-sectional tables and do not allow within-area correlation in the pooled historical regressions; Table A4 reports the corresponding clustered standard errors. Spatial standard errors use Bartlett weights and distances between representative interior points, allowing correlation across areas and years. Every regression covers 381 historical areas; observation counts include repeated years where applicable.